Land 9: Gland Spires

The endocrine system · OpenStax Ch. 17

Key ideas

Endocrine vs nervous signaling

- Endocrine glands are ductless and release hormones into blood; responses are slower to start but longer lasting than nervous responses.
- Neurons signal across synapses in milliseconds; hormones can act for minutes, hours or days.
- Only target cells with matching receptors respond to a hormone.
- Release is controlled by humoral (blood chemical), hormonal (another hormone) and neural (nerve impulse) stimuli.
- Most hormone systems use negative feedback; oxytocin in labor and nursing is a classic positive feedback loop.

Hormone classes and mechanisms

- Amines come from single amino acids (tyrosine gives epinephrine, norepinephrine, T3 and T4; tryptophan gives melatonin).
- Peptide and protein hormones (insulin, ADH, GH) are water soluble and bind membrane receptors.
- Steroids (cortisol, aldosterone, sex hormones) come from cholesterol, travel on carrier proteins and bind intracellular receptors to change gene transcription.
- Water soluble hormones use G proteins and second messengers: adenylyl cyclase makes cAMP, which activates protein kinase A; phosphodiesterase ends the signal.
- Phospholipase C makes IP3 and DAG; IP3 releases calcium from the ER, and calcium acts through calmodulin.

Receptor regulation and hormone interactions

- Downregulation: fewer receptors when hormone is persistently high, lowering sensitivity.
- Upregulation: more receptors when hormone is low, raising sensitivity.
- Permissive: one hormone is needed for another to work fully (thyroid hormone and reproductive hormones).
- Synergistic: combined effect exceeds the sum (glucagon plus epinephrine raising glucose).
- Antagonistic: opposite effects (insulin vs glucagon; calcitonin vs PTH).

Hypothalamus and pituitary

- Hypothalamic releasing and inhibiting hormones reach the anterior pituitary through the hypophyseal portal system.
- TRH to TSH (thyroid); CRH to ACTH (adrenal cortex); GnRH to FSH and LH (gonads); GHRH to GH; dopamine inhibits prolactin; somatostatin inhibits GH.
- GH acts largely through liver IGFs to build bone, muscle and cartilage; prolactin drives milk production.
- ADH and oxytocin are made in hypothalamic neurons and stored in the posterior pituitary.
- ADH inserts aquaporins in collecting ducts; oxytocin drives uterine contraction and milk ejection.

Thyroid and parathyroid

- Follicular cells make thyroglobulin; iodine is attached to its tyrosines in the colloid to form T3 and T4.
- T4 is secreted in larger amounts; T3 is more active, and tissues convert T4 to T3.
- Thyroid hormones raise basal metabolic rate, heat production and support growth and nervous system development.
- Parafollicular (C) cells release calcitonin when blood calcium is high.
- PTH from parathyroid chief cells raises calcium: osteoclast bone resorption, kidney calcium reabsorption and calcitriol formation for gut absorption.

Adrenal glands and stress

- Cortex zones from outside in: glomerulosa (aldosterone), fasciculata (cortisol), reticularis (androgens). Remember salt, sugar, sex.
- Aldosterone increases sodium reabsorption and potassium secretion, raising blood volume and pressure.
- Medulla chromaffin cells are modified sympathetic neurons releasing epinephrine and norepinephrine.
- Alarm phase: epinephrine raises heart rate, blood glucose and airflow. Resistance phase: cortisol drives gluconeogenesis and suppresses inflammation.
- Exhaustion phase follows prolonged stress and can damage organs and weaken immunity.

Pancreas and blood glucose

- Islet alpha cells make glucagon, beta cells make insulin, delta cells make somatostatin.
- Insulin lowers glucose: uptake into muscle and fat, glycogen and fat storage.
- Glucagon raises glucose: liver glycogenolysis and gluconeogenesis.
- Normal fasting glucose is about 70 to 110 mg/dL.
- Type 1 diabetes: autoimmune beta cell loss, no insulin, risk of ketoacidosis. Type 2: insulin resistance, often with obesity.

Other endocrine tissues

- Pineal melatonin rises in darkness and sets circadian rhythm.
- Testes: LH stimulates testosterone; FSH supports sperm production; inhibin limits FSH. Ovaries make estrogens and progesterone.
- Thymus thymosins promote T cell maturation.
- Heart: ANP when atria stretch. Kidney: EPO, renin, calcitriol. Adipose: leptin for satiety.
- Gut: gastrin (stomach acid), secretin (bicarbonate), CCK (enzymes and bile), ghrelin (hunger).

Endocrine disorders

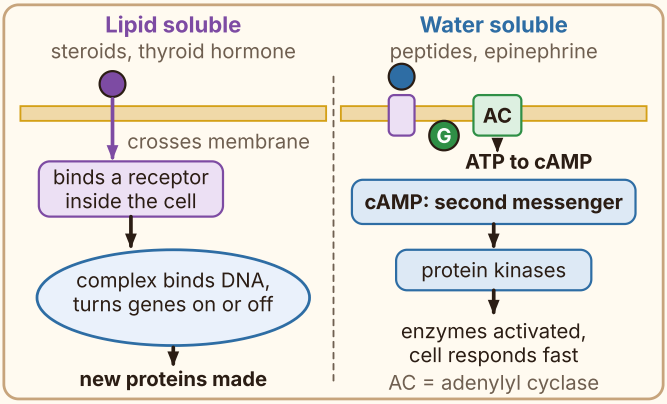
- Diabetes insipidus: ADH deficiency, large volumes of dilute urine with normal glucose.
- Hypothyroidism: fatigue, cold intolerance, weight gain. Primary shows high TSH; pituitary cause shows low TSH.
- Graves disease: antibodies stimulate TSH receptors, high T4, low TSH. Goiter often reflects iodine deficiency with high TSH.
- Cushing syndrome: cortisol excess (moon face, high glucose, thin skin). Addison disease: adrenal cortex failure (low sodium, high potassium, low pressure, darkened skin).
- GH excess: gigantism before growth plates close, acromegaly after. GH deficiency in childhood: pituitary dwarfism.

Villager wisdom

- Steroid hormones are built from cholesterol and slip right through cell membranes to bind receptors inside. Water soluble hormones such as glucagon knock at the door and rely on a second messenger such as cAMP.
- The hypothalamus is master of the master gland. It sends releasing hormones down a tiny portal blood path to the anterior pituitary.
- Insulin is the key that lets glucose into cells after a feast. Glucagon opens the liver's glycogen stores during a fast.
- Parathyroid hormone raises blood calcium by waking the osteoclasts, while calcitonin helps tuck calcium back into bone.
- Negative feedback keeps the spires in balance: when thyroid hormone is high, the hypothalamus and pituitary ease off on TRH and TSH.

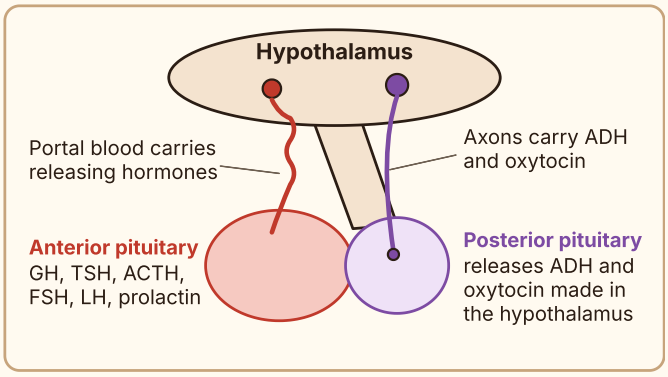
Diagrams

How hormones act on cells



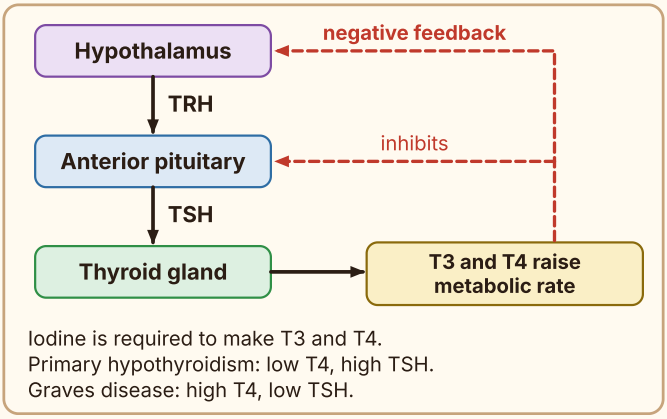
Lipid soluble hormones (steroids, thyroid hormone) cross the membrane and change gene transcription; water soluble hormones bind membrane receptors and act through second messengers such as cAMP.

Hypothalamus and pituitary



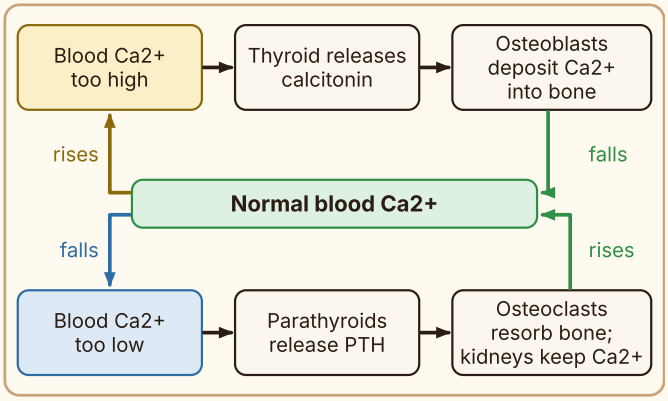
Hypothalamic releasing hormones reach the anterior pituitary through portal blood; ADH and oxytocin are made in the hypothalamus and released from the posterior pituitary.

Thyroid axis and feedback



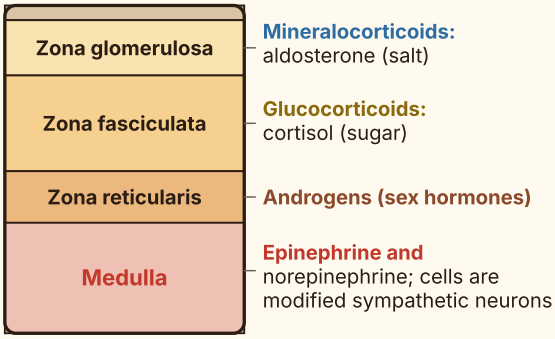
TRH stimulates TSH, which stimulates T3 and T4; rising T3 and T4 inhibit TRH and TSH, so low T4 with high TSH points to the thyroid itself.

Calcium homeostasis



PTH raises blood calcium (osteoclasts resorb bone, kidneys keep calcium and make calcitriol); calcitonin lowers it as osteoblasts deposit calcium in bone.

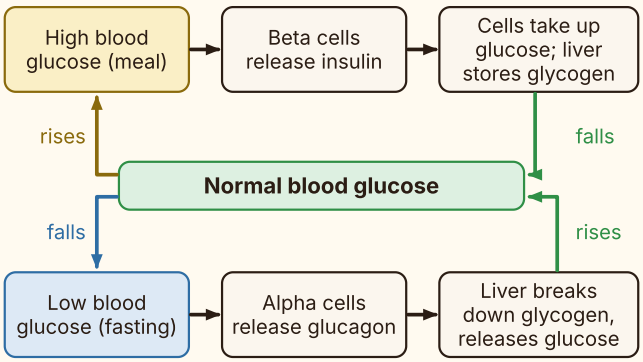
Adrenal gland zones



Too much cortisol: Cushing syndrome

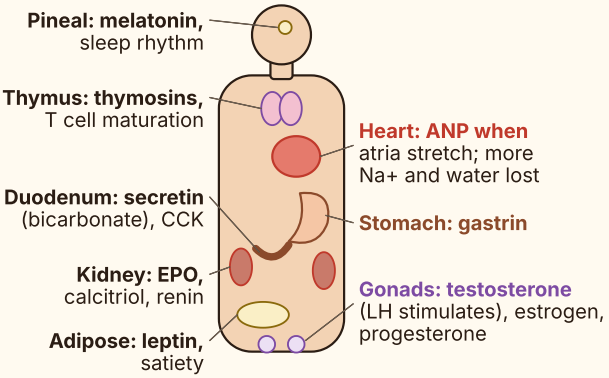
The adrenal cortex makes aldosterone (glomerulosa), cortisol (fasciculata) and androgens (reticularis); the medulla, modified sympathetic neurons, releases epinephrine and norepinephrine.

Insulin and glucagon



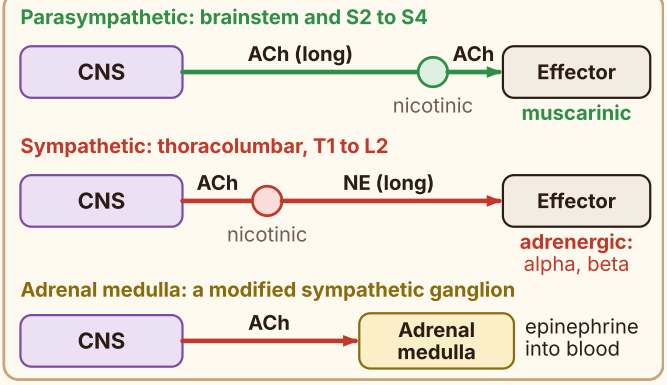
Insulin from beta cells lowers blood glucose by moving it into cells and glycogen; glucagon from alpha cells raises it by breaking down liver glycogen.

Hormones from other organs



Many organs besides the classic glands make hormones: the heart releases ANP when the atria stretch, the duodenum releases secretin, adipose tissue releases leptin, and LH stimulates testosterone from the testes.

Autonomic pathways and receptors



All preganglionic neurons release ACh onto nicotinic receptors; parasympathetic postganglionic neurons release ACh onto muscarinic receptors, while most sympathetic ones release NE onto adrenergic receptors.

Self check

1. **Parathyroid hormone raises blood calcium partly by:**
    A. Stimulating osteoblasts to build bone
    B. Increasing calcium loss in urine
    C. Stimulating osteoclasts to resorb bone
    D. Blocking vitamin D activation
2. **Which cells of the pancreatic islets secrete insulin?**
    A. Alpha cells
    B. Delta cells
    C. Beta cells
    D. Acinar cells
3. **In uncontrolled diabetes mellitus, why is urine volume high?**
    A. ADH secretion stops completely
    B. Insulin deficiency raises filtration rate
    C. Excess aldosterone causes water loss
    D. Excess glucose stays in filtrate and draws water
4. **A patient reports weight gain, fatigue, feeling cold and dry skin. Labs show low T4 and high TSH. Which condition is most likely?**
    A. Primary hypothyroidism
    B. Hyperthyroidism
    C. Hypothyroidism from pituitary failure
    D. Cushing syndrome
5. **Which anterior pituitary hormone stimulates the mammary glands to produce milk?**
    A. Oxytocin
    B. Luteinizing hormone
    C. Growth hormone
    D. Prolactin
6. **During labor, stretch of the cervix increases oxytocin release, which strengthens contractions and further stretch. This is:**
    A. Negative feedback
    B. Hormone antagonism
    C. Positive feedback
    D. Receptor downregulation
7. **A patient on high dose prednisone for months stops it abruptly and develops weakness and low blood pressure. Why?**
    A. Excess ACTH overstimulated the adrenal cortex
    B. The adrenal medulla stopped making epinephrine
    C. Suppressed ACTH left the adrenal cortex atrophied
    D. Prednisone withdrawal caused excess insulin

8. **A child is very short with normal body proportions. GH is low and thyroid tests are normal. Which condition fits best?**
    A. Congenital hypothyroidism
    B. Pituitary dwarfism
    C. Acromegaly
    D. Gigantism
9. **Reproductive hormones cannot fully promote reproductive development unless thyroid hormone is present. This is a:**
    A. Synergistic effect
    B. Antagonistic effect
    C. Permissive effect
    D. Downregulation effect
10. **Rising blood levels of T3 and T4 normally cause the hypothalamus and pituitary to:**
    A. Increase TSH secretion
    B. Decrease TRH and TSH secretion
    C. Increase TRH secretion
    D. Release more ACTH
11. **Testosterone secretion by the interstitial cells of the testes is directly stimulated by:**
    A. Follicle stimulating hormone (FSH)
    B. Luteinizing hormone (LH)
    C. Prolactin
    D. Gonadotropin releasing hormone
12. **Aldosterone acts on the kidney tubules to:**
    A. Increase potassium reabsorption and sodium loss
    B. Increase sodium reabsorption and potassium secretion
    C. Increase water excretion and lower blood volume
    D. Increase calcium reabsorption only

Answer key

1. **C.** Stimulating osteoclasts to resorb bone. PTH activates osteoclasts to break down bone matrix and release calcium, and it also boosts kidney reabsorption and calcitriol production. Stimulating osteoblasts would pull calcium out of the blood into bone, lowering it.
2. **C.** Beta cells. Beta cells release insulin when blood glucose rises. Alpha cells secrete glucagon, the hormone with the opposite effect, and acinar cells make digestive enzymes rather than hormones.
3. **D.** Excess glucose stays in filtrate and draws water. When filtered glucose exceeds what the tubules can reabsorb, it remains in the filtrate and osmotically holds water, causing polyuria. Loss of ADH causes diabetes insipidus, where urine contains no glucose.
4. **A.** Primary hypothyroidism. A failing thyroid makes too little T4, and the pituitary responds with high TSH, which points to a problem in the thyroid itself. If the pituitary had failed, TSH would be low along with T4.
5. **D.** Prolactin. Prolactin from the anterior pituitary drives milk production. Oxytocin is tempting, but it comes from the posterior pituitary and causes milk ejection (letdown), not production.

6. **C.** Positive feedback. The response amplifies the original stimulus until delivery ends the loop, which defines positive feedback. Negative feedback would reduce the stimulus and return the system toward a set point.
7. **C.** Suppressed ACTH left the adrenal cortex atrophied. Long term glucocorticoids suppress CRH and ACTH by negative feedback, so the adrenal cortex shrinks and cannot make enough cortisol when the drug stops. ACTH is low, not high, during and right after such therapy.
8. **B.** Pituitary dwarfism. GH deficiency in childhood produces short stature with normal proportions and can be treated with GH. Congenital hypothyroidism is ruled out by normal thyroid tests and typically causes disproportionate growth and developmental delay.
9. **C.** Permissive effect. In a permissive effect one hormone must be present for another to act fully, often by increasing its receptors. Synergism means two hormones produce a combined effect greater than their sum, which is not what is described.
10. **B.** Decrease TRH and TSH secretion. This is negative feedback: high thyroid hormone suppresses TRH and TSH so the thyroid slows down. Increasing TSH would push hormone levels even higher, which is positive feedback and would not maintain balance.
11. **B.** Luteinizing hormone (LH). LH acts on interstitial (Leydig) cells to trigger testosterone release. FSH is tempting, but it acts on sustentacular cells to support sperm production.
12. **B.** Increase sodium reabsorption and potassium secretion. Aldosterone retains sodium (and water follows) while promoting potassium excretion, raising blood volume and pressure. The reversed ion pattern is the opposite of its action.

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